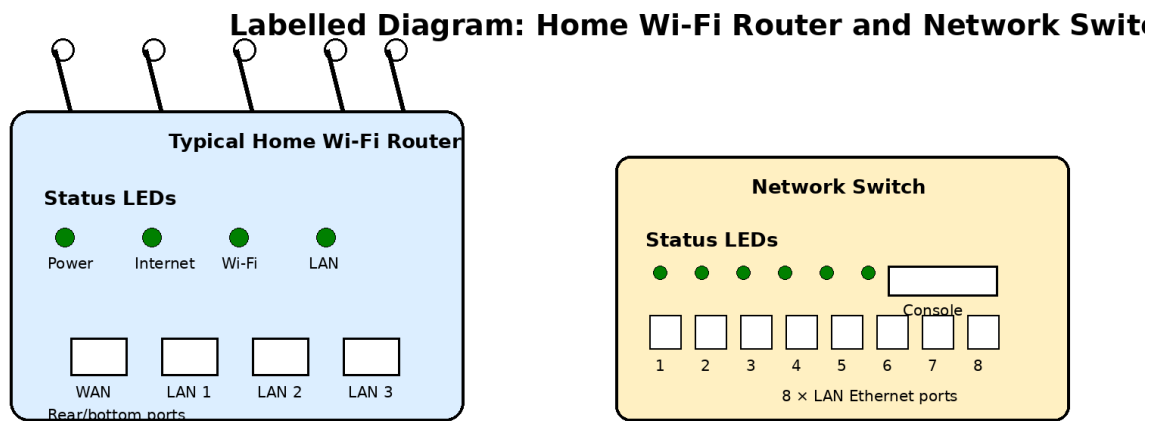


Assignment:-2

Task 1:- Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

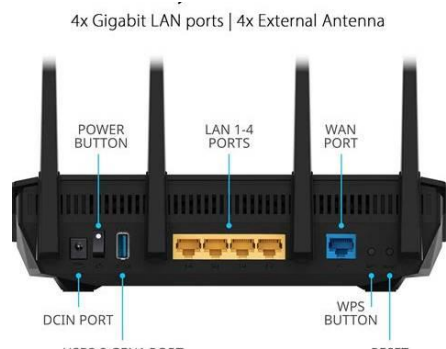


WAN = Internet input LAN = Local wired connection Console = Device
Status LEDs show power, internet/link, activity, or port status.

Task 2:- List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

No.	Router	Switch
1. Main Function	Connects different networks, such as a home network to the Internet.	Connects multiple devices within the same local network (LAN).
2. Address Used	Uses IP addresses to route packets between networks.	Uses MAC addresses to forward frames to the correct device.
3. Network Traffic	Transfers traffic between different networks or subnets.	Transfers traffic between devices in the same LAN.

Task 3:- Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



Task 4:- Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices..

Scenario: Small Gaming Café Network

I am setting up a small gaming café for IPL streaming and multiplayer gaming. First, I would connect the **ISP internet line to the router**. The router would manage the internet connection, assign IP addresses, and provide network security.

Then, I would connect one of the router's **LAN ports to a network switch** using an Ethernet cable. Finally, I would connect all **8 gaming PCs to the switch** using separate Ethernet cables.

Connection layout:

ISP → Router → Switch → 8 Gaming PCs

Justification:

The router is required to connect the café network to the internet and manage traffic between networks. The switch is useful because it provides multiple Ethernet ports and allows all 8 gaming PCs to communicate efficiently over a wired LAN, which is suitable for multiplayer gaming and stable IPL streaming.